IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

EEE/EIE PART III/IV: MEng, BEng and ACGI

DIGITAL SIGNAL PROCESSING

Thursday, 7 December 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P.T. Stathaki
Second Marker(s) : W. Dai

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1. (a) Consider the causal, real-coefficient transfer function:

$$H_M(z) = \frac{\sum_{n=0}^{n=M} b[n]z^{-n}}{\sum_{n=0}^{n=M} a[n]z^{-n}}, M > 0$$

(i) Give the condition that must hold for $H_M(z)$ to be an allpass filter. Justify you answer. **[3]**

(ii) Comment on the locations of the zeros of a causal, real, stable, allpass filter's transfer function within the z −plane, with respect to the unit circle and the poles. Justify your answer. **[3]**

(b) Consider a first-order, causal, real, stable, allpass transfer function $H(z)$ given as:

$$H(z) = \frac{p + z^{-1}}{1 + pz^{-1}}$$

(i) Determine the impulse response $h[n]$ for $n \geq 0$, associated with $H(z)$, if $h[n] = 0$ for $n < 0$. **[5]**

(ii) Find the phase response associated with $H(z)$. **[4]**

(iii) Determine the expression for $|H(z)|^2 - 1$. **[3]**

(iv) Find for which values of z the condition $|H(z)|^2 < 1$ holds. **[3]**

(c) What is a major drawback in designing an all-pole IIR linear-phase transfer function? Justify your answer. **[4]**

2. (a) Show that an antisymmetric linear-phase Finite Impulse Response (FIR) transfer function $H(z)$ of odd length N can be expressed as follows:

$$H(z) = z^{-\left(\frac{N-1}{2}\right)}\left(\sum_{n=1}^{\frac{N-1}{2}} h\left[\frac{N-1}{2} - n\right](z^n - z^{-n})\right) \qquad (1)$$

[8]

(b) By using the relation

$$U_r\left(\frac{z+z^{-1}}{2}\right) = \frac{z^{r+1} - z^{-(r+1)}}{z - z^{-1}}$$

where $U_r(x)$ is the r −th order **Chebyshev Polynomial of the Second Kind** in x, express $H(z)$ of (a) above in the alternative form

$$H(z) = z^{-\left(\frac{N-1}{2}\right)}(z - z^{-1}) \sum_{n=0}^{M} a[n]\left(\frac{z+z^{-1}}{2}\right)^n \qquad (2)$$

in the case where $N = 9$. Determine the relation between $a[n]$ and $h[n]$ for $N = 9$. You can easily determine the value of the parameter M as a function of N, by ensuring that the orders of the polynomials of equations (1) and (2) are equal.

The **Chebyshev Polynomials of the Second Kind** satisfy the following recursive relationship:

$U_0(x) = 1$
$U_1(x) = 2x$
$U_r(x) = 2xU_{r-1}(x) - U_{r-2}(x), r \geq 2$

[8]

(c) Develop a realization of $H(z)$ based on equation (2) above in the form of **Figure 2.1** below, where $F_1(z^{-1})$ and $F_2(z^{-1})$ are causal filters. Determine the form of $F_1(z^{-1})$ and $F_2(z^{-1})$. The triangular sign indicates multiplication with the parameter shown next to it.
[**Hint:** To solve this part, it is convenient to write equation (2) as a function of M only instead of N only or both M and N]. **[9]**

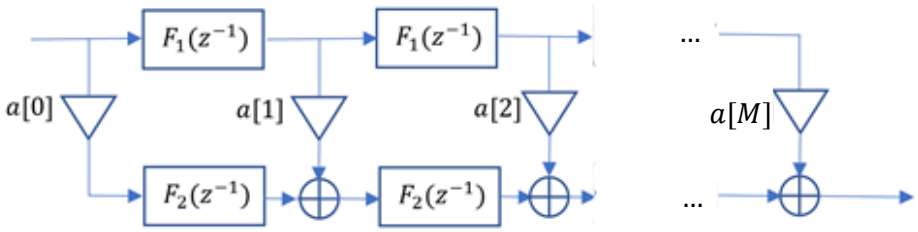


Figure 2.1

3. (a) The bilinear transformation from the s −plane to the z −plane is given by

$$s = \frac{1 - z^{-1}}{1 + z^{-1}}$$

(i) Illustrate, employing mathematical relationships, how the bilinear transformation maps every point $s = \sigma + j\Omega$ in the s −plane to the z −plane. **[5]**

(ii) Prove that the relationship between the continuous-time angular frequency and the discrete-time angular frequency is non-linear. **[5]**

(b) A given real-coefficient, digital IIR lowpass filter has a rational transfer function $H_L(z)$ and a cutoff frequency ω_c. The transfer function of $H_L(z)$ is transformed by replacing z by $F(z) = \frac{az+b}{cz+d}$ to a real highpass rational transfer function $H_H(z) = H_L(F(z))$ of the same order as $H_L(z)$ but different cutoff frequency. By using the constraints $H_H(e^{j\pi}) = H_L(e^{j0}) = 1$ and $H_H(e^{j0}) = H_L(e^{j\pi}) = 0$, derive relationships among the parameters a, b, c, d and explain what type of filter is $F(z)$. **[5]**

(c) Consider the two LTI causal digital filters with impulse responses given by:

$$h_A[n] = 0.5\delta[n] - \delta[n-1] + 0.5\delta[n-2]$$
$$h_B[n] = 0.25\delta[n] + 0.5\delta[n-1] + 0.25\delta[n-2]$$

Explain the type of filters $h_A[n]$ and $h_B[n]$ (lowpass etc.), by using the following two approaches:

(i) Practical approach. In this approach you will apply the filters to the input signal $u[n]$, with $u[n]$ the discrete unit step function and observe the effects they have on $u[n]$. **[5]**

(ii) Mathematical approach. In this approach you must find and sketch the amplitude response of the two filters. **[5]**

4. (a) Examine the multirate configuration illustrated in **Figure 4.1** below. In this setup, $H_0(z), H_1(z)$ and $H_2(z)$ are lowpass, bandpass, and highpass filters, each possessing ideal, zero-phase characteristics and real coefficients. Their corresponding frequency responses are outlined as follows:

$$H_0(e^{j\omega}) = u(\omega) - u(\omega - \frac{\pi}{3})$$

$$H_1(e^{j\omega}) = u(\omega - \frac{\pi}{3}) - u(\omega - \frac{2\pi}{3})$$

$$H_2(e^{j\omega}) = u(\omega - \frac{2\pi}{3}) - u(\omega - \pi)$$

The function $u(\omega)$ is the unit step function, defined as

$$u(\omega) = \begin{cases} 1 & \omega \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

If the input is a real sequence with a Discrete Time Fourier Transform

$$X(e^{j\omega}) = (-\frac{3\omega^2}{\pi} + \frac{\pi}{3})\left(u(\omega) - u\left(\omega - \frac{\pi}{3}\right)\right)$$

sketch the Discrete Time Fourier Transform of the outputs $y_0[n], y_1[n]$ and $y_2[n]$.
It's important to note that all frequency domain representations mentioned are limited to the range $[0, \pi]$. **[15]**

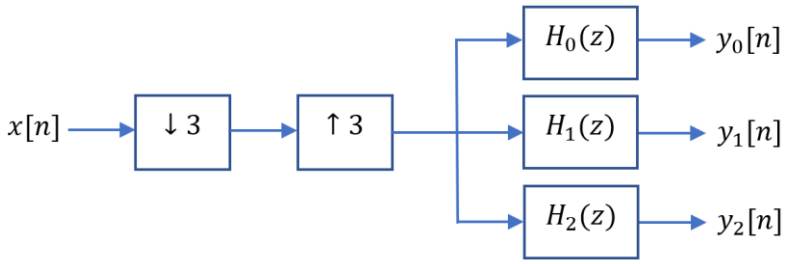


Figure 4.1

(b) Analyse the structure of **Figure 4.2** below and determine its input-output relations. **[10]**

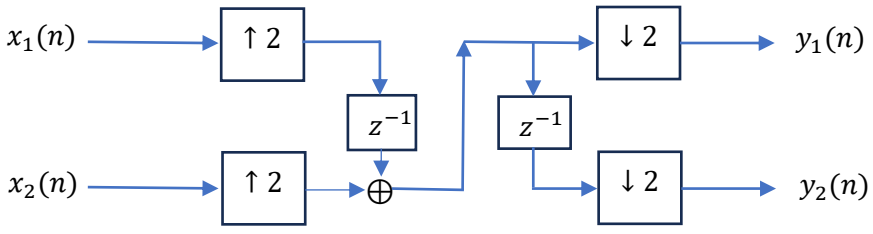


Figure 4.2